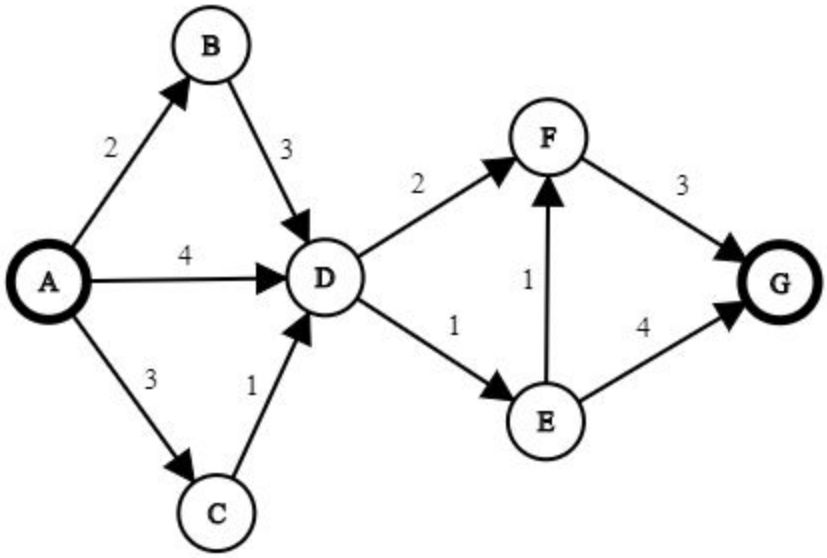
1. If *h1* and *h2* are two admissible heuristics, then which of the following are guaranteed to be admissible?
    a. **min(h1,h2)**
    b. **max(h1,h2)**
    c. **(h1+h2)/2**
    d. **(h1.h2)^0.5**

    Let real cost to goal for each node be given by function cost(n).
    We know, h1(n)≤cost(n) and h2(n)≤cost(n) for every node n. (h1, h2 are admissible)
    Therefore, max(h1(n),h2(n)) ≤ cost(n) for every node n.
    Hence max(h1,h2) is admissible.
    Similarly, this can be shown for others.

2. If *h1* and *h2* are two admissible heuristics, then which of the following will give a heuristic that dominates both *h1* and *h2*?
    a. min(h1,h2)
    b. max(h1,h2)
    c. (h1+h2)/2
    d. (h1.h2)^0.5

    A heuristic h1 is said to dominate another heuristic h2 if for every node h2(n)≤h1(n).
    Using this definition it is easy to see that max(h1,h2) dominates both h1 and h2.

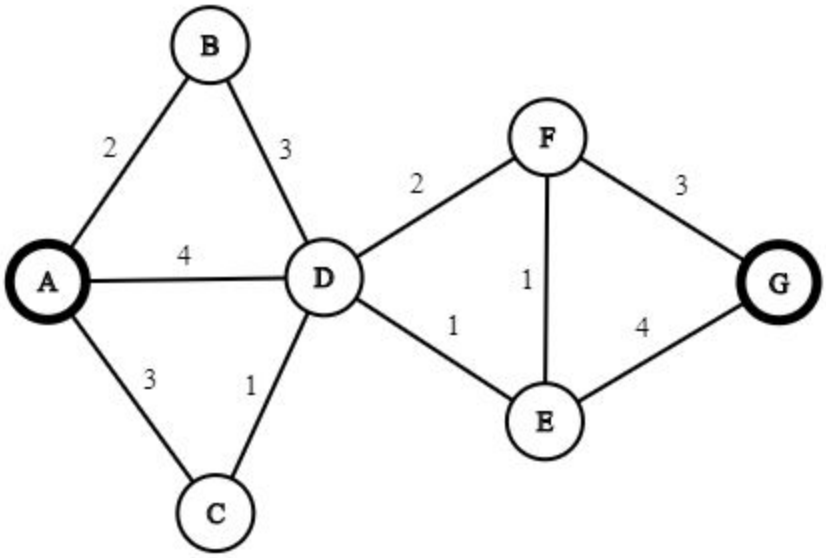
3. Consider the following directed graph in which we are searching from start state A to goal state G. The number over each edge is the transition cost. Additionally, we are given a heuristic function h as follows {h(A)=7, h(B)=5, h(C)=6, h(D)=4, h(E)=3, h(F)=3, h(G)=0}.
    The given heuristic function is:



a. Admissible as well as consistent
b. Admissible but not consistent
c. Consistent but not admissible
d. Neither admissible nor consistent

The given heuristic is admissible since for every node the minimum cost to goal is higher than that given by heuristic.
However, the given heuristic is not consistent since
h(D)+Cost(C,D) ≤ h(C)

4. Consider the following undirected graph in which we are searching from start state A to goal state G. The number over each edge is the transition cost. Additionally, we are given a heuristic function h as follows {h(A) = 7, h(B) = 5, h(C) = 6, h(D) = 4, h(E) = 3, h(F) = 3, h(G) = 0}. Find the sequence of nodes explored (i.e., the sequence of removals from its frontier) by A* graph search algorithm. Assume that, in case of ties, the search procedure uses a lexicographical order for tie-breaking. Write the answer as a string with no spaces. For example, if the order of exploration is A followed by B followed by C followed by D then write ABCD.



Ans. ABDECFG

Iteration	g(n)+h(n)	Frontier	Node explored
1	{'A': 7} A 7	['A']	A
2	{'B': 7, 'C': 9, 'D': 8}	['B']	B
3	{'C': 9, 'D': 8}	['D']	D
4	{'C': 9, 'E': 8, 'F': 9}	['E']	E
5	{'C': 9, 'F': 9, 'G': 9}	['C', 'F', 'G']	C
6	{'F': 9, 'G': 9}	['F', 'G']	F
7	{'G': 9}	['G']	G

5. Which of the following algorithms is/are guaranteed to give an optimal solution?
    a. Greedy Best First Search
    b. A* with zero heuristic
    c. A* with consistent heuristic
    d. Depth First Search

    A* with consistent heuristic is optimal. Zero heuristic is also consistent. DFS and Greedy Best First are not optimal.

6. Consider an infinite search space $\mathbb{Z} \times \mathbb{Z}$. The start state is at (0, 0) and the goal state is at (g_x, g_y). Given that the agent can move from a state (x, y) to either of {(x+1, y), (x-1, y), (x, y+1), (x, y-1)} with a unit step cost, find the number of nodes explored using A* tree search with Manhattan distance heuristic for $(g_x, g_y) = (6, 7)$. Assume no duplicate detection.
    Ans: 14

    Consider an infinite grid in the cartesian plane. The user starts at the origin and can move to any neighboring integer point. The Manhattan distance can be observed to be the perfect estimate of the shortest distance from current state to the goal state. Therefore, the A* exploration node order will be exactly the shortest path from the origin to the goal state. Note that there may be many such paths (no. of permutations of 6 ups and 7 rights) but all of them will contain 14 nodes.

7. Which of the following statement(s) is/are correct?
    a. A* graph search expands no more nodes than uniform-cost graph search.

b. **Uniform cost search is a special case of best-first tree search.**
c. A* graph search expands no more nodes than depth-first graph search.
d. Weighted A* implementation does not need a priority queue.

a. The heuristic helps to guide the search and reduce the number of nodes expanded. In the extreme case where the heuristic function returns zero for every state, the no. of nodes explored will be the same.
b. Best-first search with f(n) = g(n) gives the Uniform cost search algorithm.
c. Depth first graph search in the best case can be O(n) whereas the A* graph search can still be exponential.
d. Weighted A* needs a priority queue.

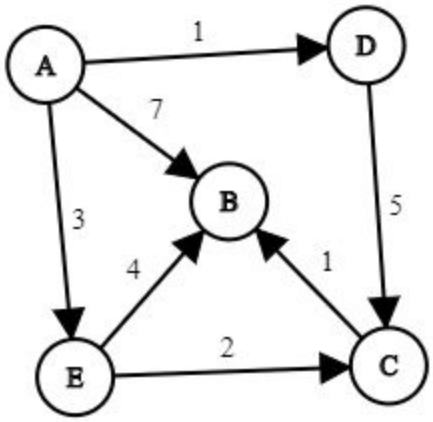
8. Consider the A* tree search algorithm with $f(n) = (2 + x).g(n) + (x^2).h(n)$ where g(n) is the cost from the start state to state n and h(n) is an admissible estimate of the cost from state n to the goal state. For how many integer values of x is the algorithm guaranteed to be optimal?
Ans: 4

A* tree search algorithm with f(n) = w1.g(n) + w2.h(n) = w1.(g(n) + (w2/w1).h(n)) gives the optimal solution when w1 >= w2 as it ensures that the heuristic value is not overestimating. Solving for (2 + x >= x^2) gives x = -1, 0, 1, 2 as the integer solutions.

9. Consider a heuristic function $h_2(n) = 3.h_1(n)$ where $h_1(n)$ is admissible. Which of the following statement(s) is/are correct?
   a. $h_2(n)$ is admissible.
   b. A* graph search with h_2 will return the optimal path.
   c. **A* tree search with h_2 will return a path at most thrice as long as the optimal path.**
   d. **$h_2(n) + 1$ is guaranteed to be inadmissible for any h_1.**

a. $h_2(n)$ might overestimate the cost.
b. Since $h_2(n)$ is not admissible A* graph search will not return the optimal path.
c. In A* tree search we have that as long as the optimal path to the goal has not been found, a prefix of this optimal path has to be on the fringe. Hence, if a non-optimal solution is found, then at the time of popping the non-optimal path from the fringe, a path that is a prefix of the optimal path to the goal is sitting on the fringe. The cost g' of a non-optimal solution when popped is its f-cost. The prefix of the optimal path to the goal has an f-cost of $g + h_2 = g + 3h_1 \leq 3(g + h_1) \leq 3C^*$, with C* the optimal cost to the goal. Hence we have that $g' \leq 3C^*$ since $g' < g + h_2$ as the non-optimal path was chosen over the optimal path from the fringe.
d. $h_2(goal) + 1 = 1$ since $h_1(goal) = 0$. Therefore, $h_2(n)$ is always inadmissible.

10. Consider the following directed graph in which we are searching from start state A to goal state B. The number over each edge is the transition cost. Additionally, we are given a heuristic function h as follows {h(A) = 5, h(B) = 0, h(C) = 1, h(D) = 6, h(E) = 2}. Find the sequence of nodes explored by Depth-First Branch and Bound Search Algorithm. A node is explored further only if $g(n_k) + h(n_k) <$ bound. Assume that the search procedure uses a lexicographical order for exploration/tie-breaking. Write the answer as a string with no spaces. For example, if the order of exploration is A followed by B followed by C followed by D then write ABCD.



Ans: ABDEBCB

Initially, the bound is infinity and A is explored.
Then, checking lexicographically, B is chosen and bound is set to 7.
Then, exploring D and we find g(D) + h(D) = bound hence we don't explore it "further".
Then, explore E and we find that g(E) + h(E) < bound hence we explore it further.
Then, explore B and we find that g(B) + h(B) = bound hence we don't explore it "further".
Then, explore C and we find that g(C) + h(C) < bound hence we explore C further.
Then, we explore B and we are done.

The word "further" might have caused a confusion here and some of you might have come up with different answers. We'll try highlighting these words in the upcoming assignments, in fact prefer avoiding these kinds of ambiguities.